12. _____ is added to 24^2 to get 25^2.

(A) 4^2 (B) 5^2 (C) 6^2 (D) 7^2

13. $\sqrt{48}$ is approximately equal to _____.

(A) 5 (B) 6 (C) 7 (D) 8

14. $\sqrt{128} - \sqrt{98} + \sqrt{18}$ = _____.

(A) $\sqrt{2}$ (B) $\sqrt{8}$ (C) $\sqrt{48}$ (D) $\sqrt{32}$

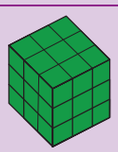
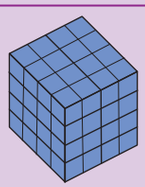
15. The number of digits in the square root of 123454321 is _____.

(A) 4 (B) 5 (C) 6 (D) 7

1.8 Cubes and Cube Roots

If you multiply a number by itself and then by itself again, the result is a cube number. This means that a cube number is a number that is the product of three identical numbers. If n is a number, its cube is represented by n^3.

Cube numbers can be represented visually as 3D cubes comprising of single unit cubes. Cube numbers are also called as perfect **cubes**. The perfect cubes of natural numbers are 1, 8, 27, 64, 125, 216, ... and so on.

Geometrical Representation	Product Representation	Notation	Perfect cube
	$1 \times 1 \times 1$	1^3	1
	$2 \times 2 \times 2$	2^3	8
	$3 \times 3 \times 3$	3^3	27
	$4 \times 4 \times 4$	4^3	64



Ramanujan Number - $1729 = 12^3+1^3 = 10^3+9^3$

Once Professor Hardy went to see Ramanujan when he was ill at Putney, riding in taxi cab number 1729 and said that the number seemed a dull one, and hoped it was not an unfavourable omen. "No," replied Ramanujan and he completed saying "It is a very interesting number. Infact, it is the smallest number expressible as the sum of two cubes in two different ways." 4104, 13832, 20683 are a few more examples of Ramanujan-Hardy numbers.

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